GURMILAN SINGH

Full-Stack Web Developer | Software Engineer | React · JavaScript · Node.js · SQL · C#/.NET

Modena, Italy · gurmilans01@gmail.com

linkedin.com/in/gurmilan-singh-017b28284 · github.com/gurmilans-dev · gurmilansingh.com

PROFESSIONAL SUMMARY

Final-year Computer Engineering student and software developer with 2+ years of professional experience building and maintaining production full-stack applications. Primary developer of an internal business platform, with end-to-end responsibility spanning frontend and backend development, relational database design, authentication, deployment, maintenance, and user support. Experienced with C#/.NET, React, Node.js, SQL, automation, infrastructure support, and AI/API integrations.

PROFESSIONAL EXPERIENCE

Software Developer — C#/.NET & Full-Stack Web

Phema S.r.l. · Modena, Italy · 09/2023 – Present

React · JavaScript · Material UI · Node.js · Express · SQL Server · Jest · C#/.NET · systems administration

- Serve as primary developer of a production full-stack internal business application supporting shared operational workflows and role-based access.
- Develop and maintain modules for operational records, project coordination, scheduling, time management, reporting and administrative workflows.
- Own end-to-end delivery across requirements analysis, frontend and backend development, relational database design, authentication, testing, deployment, maintenance and user support.
- Helped replace disconnected legacy tools and manual processes with a centralized web application and relational data model.
- Continuously deliver production features in response to internal requirements, covering implementation, testing, release, defect resolution and ongoing support.
- Contribute to the development and maintenance of the company's commercial C#/.NET software.
- Built a C#/.NET client-server licence-management component supporting licence validation and controlled concurrent use.
- Support the software's server and virtualized operating environment, diagnosing application and infrastructure issues.

SELECTED PROJECTS

Opssemble — AI-Assisted Event Operations Platform

React 18 · Vite · Node.js · Fastify · AI APIs · JWT · REST API · Nodemailer · MySQL · Google OAuth 2.0

- Built **Opssemble**, a full-stack event operations platform designed to help teams coordinate complex events, manage operational areas, track tasks and issues, monitor readiness, and give on-site staff controlled access to the information they need.
- Designed a modular backend architecture supporting event planning, team coordination, task management, operational exceptions, readiness monitoring, reporting, activity history, staff access, templates, and administrative workflows.
- Implemented secure account management with email verification, password recovery, social authentication, protected browser sessions, login abuse prevention, and role-based access controls.
- Created a four-level permission model for owners, coordinators, operators, and viewers, ensuring that users can only access or modify resources within their assigned responsibilities.
- Developed revocable QR-based access for on-site personnel, allowing temporary and scoped access to specific operational areas without exposing the broader event workspace.
- Integrated an AI-assisted workflow for drafting tasks and issues, generating event structures, identifying planning gaps, explaining readiness scores, summarizing blockers, and preparing post-event reports. AI-generated suggestions remain subject to user review and existing authorization rules.
- Added production-oriented capabilities including audit trails, database migrations, transactional notifications, readiness scoring, downloadable reports, secure file validation, request throttling, input validation, and defensive API security controls.

Gurmat Saanj — Full-Stack Web App with Remote Control System

React · JavaScript · Vite · Node.js · Express · BaniDB · Web Speech API ·

- Built a full-stack app for real-time Punjabi speech recognition with **fuzzy Shabad search**, confidence scoring, automatic line tracking, bilingual display, configurable projector display.
- Implemented mobile session access through **QR-code scanning** or a four-digit connection code, allowing authorised users to **remotely control** the projector presentation from a phone.
- Implemented **real-time synchronisation** between the main application, projector view, and connected mobile devices.
- Hardened the API with **Helmet, CORS, rate limiting, and input validation**.

Azure VM Provisioning Automation

Bash · Azure CLI · Linux (Ubuntu) · SSH · University project

- Wrote **Bash (.sh) scripts to automate provisioning of Ubuntu virtual machines on Microsoft Azure** over SSH, replacing a manual setup process with a repeatable, scripted workflow.

Smart Bin — IoT Prototype

Arduino

- Prototyped a smart waste bin with sensor- and motor-driven automatic lid opening that transmits sensor data to a companion mobile app.

SIRD Epidemic Simulator — Java Desktop Application

Java 17 · JavaFX · Maven · FXML · Concurrency

- Developed an agent-based epidemic simulator with configurable population, infection radius and probability, recovery time, and mortality rate, including pause, resume, quarantine and reset controls.
- Implemented a background simulation engine, real-time S/I/R/D visualisation, moving agents with boundary collision handling, and CSV data export using a model–engine–controller architecture.

TECHNICAL SKILLS

Languages: C#, TypeScript, JavaScript, Java, Python, SQL, C, Bash
Backend: ASP.NET Core, .NET, Entity Framework Core, Node.js, Express, Fastify, REST APIs, JWT, RBAC
Frontend: React, Vite, HTML5, CSS3, Material UI, React Router, TanStack Query
Databases: SQL Server, MySQL, relational database design, migrations
Cloud/DevOps: Azure, Azure CLI, Docker, Linux, SSH, VMware ESXi, Git, GitHub, Postman
AI & Integrations: LLM API integration, Ollama, Groq, Google OAuth 2.0, Web Speech API

EDUCATION

BSc in Software Engineering - Laurea in Ingegneria Informatica
University of Modena and Reggio Emilia (UNIMORE) · 2023 – Present · Expected December 2026

Final year. Core areas: programming, algorithms, data structures, databases, networks, operating systems, software engineering, web technologies, and machine learning / AI.

Diploma in Computer Science and Telecommunications
ITIS Enrico Fermi, Modena · 2018 – 2023

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Cisco Certificate — Cisco Networking Academy
- The Web Developer Bootcamp (74 hrs) — Udemy
- Self-directed online coursework in C#, C, Java, Python, and React

LANGUAGES

- Italian: Native · Punjabi: Native · English: C1 Advanced · Hindi: Advanced

Additional information: Italian driving licence, category B

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